

PROBABILITY THEORY

mm 3

Problem 3.1 (problem 4.27 from Sheldon Ross, 3rd ed.)

The density function of X is given by

$$f(x) = \begin{cases} a + bx^2 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

If $E[X] = 3/5$, find a and b .

Problem 3.2 (problem 4.32 from Sheldon Ross, 3rd ed.)

If $E[X] = 2$ and $E[X^2] = 8$, calculate

- (a) $E[(2 + 4X)^2]$
- (b) $E[X^2 + (X + 1)^2]$

Problem 3.3 (problem 4.45 from Sheldon Ross, 3rd ed.)

A product is classified according to the number of defects it contains and the factory that produces it. Let X_1 and X_2 be the random variables that represent the number of defects per unit (taking the possible values of 0, 1, 2, or 3) and the factory number (taking on possible values 1 or 2), respectively. The entries in the table represent the joint probability mass function of a random chosen product:

$$P(0, 1) = 1/8; P(0, 2) = 1/16; P(1, 1) = 1/16; P(1, 2) = 1/16;$$

$$P(2, 1) = 3/16; P(2, 2) = 1/8; P(3, 1) = 1/8; P(3, 2) = 1/4$$

- (a) Find the marginal probability distributions of X_1 and X_2 .
- (b) Find $E[X_1]$, $E[X_2]$, $Var[X_1]$, $Var[X_2]$, and $Cov(X_1, X_2)$.

Problem 3.4 (problem 4.53 from Sheldon Ross, 3rd ed.)

Suppose that X has density function

$$f(x) = e^{-x}, x > 0$$

Compute the moment generating function of X and use your results to determine its means and variance. Check your answer for the mean by a direct calculation.

Problem 3.5

A cdf of a r.v. X is

2

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x/4 & 0 < x \leq 4 \\ 1 & x > 4 \end{cases}$$

Find $E[X]$.